

GoodHive — Technical Challenge

Candidate: Aswin Kumar

Role: Junior Blockchain Developer

Issued by: Jubayer Juhan — GoodHive

Deadline: 5 days from receipt

Task: Build a Certificate Registry Smart Contract

Deploy a Solidity smart contract on **Ethereum Sepolia testnet** that allows an institution to issue, verify, and revoke certificates on-chain.

Requirements

1. Issue a Certificate

```
function issueCertificate(
    address recipient,
    string calldata courseName,
    string calldata ipfsHash
) external onlyOwner returns (uint256 certificateId)
```

- Only the contract owner can issue certificates
 - Each certificate must have a unique auto-incrementing ID
 - Store: recipient address, course name, IPFS hash, timestamp, and issuer address
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2. Verify a Certificate

```
function verifyCertificate(uint256 certificateId)
    external view
    returns (address recipient, string memory courseName, uint256 issuedAt, bool
    isValid)
```

- Anyone can verify a certificate by its ID
 - A certificate must return `isValid = false` if it has been revoked
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3. Revoke a Certificate

```
function revokeCertificate(uint256 certificateId) external onlyOwner
```

- Only the owner can revoke a certificate
 - Revoked certificates must reflect `isValid = false` in the verify function
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4. Events

Your contract must emit the following events:

```
event CertificateIssued(uint256 indexed certificateId, address indexed recipient,  
string courseName);  
event CertificateRevoked(uint256 indexed certificateId);
```

Bonus (Optional)

- Add a function to retrieve all certificate IDs issued to a specific recipient address
- Write tests using Hardhat or Foundry covering: issue, verify, and revoke

Deliverables

Please submit a GitHub repository containing:

File	Description
contracts/CertificateRegistry.sol	Your smart contract
README.md	Deployed contract address on Sepolia + brief explanation of your design choices
test/ (bonus)	Test file if you wrote tests

Submission

Send your GitHub repository link to: davidjuhan23@gmail.com

Subject: [Blockchain Challenge] Aswin Kumar – Submission

Questions? Reply to this email.

Good luck — we look forward to reviewing your work.

Jubayer Juhan, GoodHive